

Pavithra Venkat Sridhar

Evanston, IL | pavithravenkatsridhar2027@u.northwestern.edu | linkedin.com/in/pavi-venk | pavi-venk.github.io/portfolio

EDUCATION

Northwestern University, USA | Bachelor of Science in Mechanical Engineering

Sept. 2023 - Present

Robotics concentration, Segal Design Certificate

Coursework: Mechatronics, Mechanical Design, Dynamic Systems, Feedback Systems

Awards & Honors: Dean's List, Tau Beta Pi, Pi Tau Sigma, RISE Germany scholarship

RELEVANT EXPERIENCE

Hyphen

Jun. 2026 - Sept. 2026

Systems and Test Engineering Intern

San Jose, CA

- Designed and manufactured a 48 sq. ft automated temperature chamber ceiling for Makeline rigorous environmental testing, replacing manual venting and tightening temperature control by 2.5x
- Engineered a dispenser shaft adapter for in-house dispense testing and user-facing deployment
- Implemented initial controls for a 6-axis robot arm to support end effector integration and testing

Northwestern University Space Technology and Rocketry Society

Sept. 2023 - Present

Chief Engineer (Incoming 2026-2027)

Evanston, IL

Launch Vehicle Lead

- Led design for the International Rocketry Engineering Competition 2026 team rocket, 10k ft altitude category
- Developed modular fin system for Student Launch 2025, presenting design reviews to NASA technical experts
- Fabricated carbon fiber wound coupling element to accommodate 20% larger payload airframe diameter
- Earned NAR Level 2 High Power Rocketry certification in 2025 with J-class impulse motor

German Aerospace Center

Jun. 2025 - Aug. 2025

Engineering Research Intern

Stuttgart, Germany

- Conducted thermal analysis of 18+ self-prepared thermoplastic samples with Differential Scanning Calorimetry
- Analyzed user-influenced variability to improve integral bound-selection and reduce enthalpy uncertainty by 15%
- Automated analysis of 300+ thermal heating curves using Python (NumPy, Pandas, SciPy, Matplotlib libraries)

Meemo

Jul. 2025 - Sept. 2025

Industrial Design Intern

Remote

- Engineered two adjustable strap configurations to support ergonomic fit of distance trackers for parent-child users
- Modified prototype device housing shell using SolidWorks to integrate haptics motor, speaker and PCB
- Applied Design for Assembly principles to produce CAD and assembly plans for prototype manufacturing

Center for Robotics and Biosystems, Swarm Robotics Lab

Mar. 2024 - May 2026

Undergraduate Research Assistant

Evanston, IL

- Developed OptiTrack visual-based program in Python/C supporting multi-landing sequence for 50 aerial robots
- Conducted flight experiments, tuning PID/LQR controllers to improve stability and position accuracy to 2 cm
- Designed and prototyped lightweight PLA mechanical housings of under 2.3 grams to mount onboard electronics

Activities: NU Society of Women Engineers Co-President (2025-2026), Dearborn Observatory Tour Guide

SKILLS

Programming & Controls: Python, C, MATLAB/Simulink, PID/LQR control, Linux (CLI), I2C, SPI, CAN, UART

CAD & Design: OnShape, SolidWorks, Siemens NX, GD&T, DFM/DFA

Prototyping & Testing: 3D printing (FDM, SLS), CNC milling, composite fabrication, UTM testing, DSC analysis, OptiTrack motion capture

Languages: English (Native), French (Fluent), German (Beginner)